MEMSAD: Gradient-Coupled Anomaly Detection for Memory Poisoning in Retrieval-Augmented Agents

Anonymous Author(s)
Affiliation
Address
email

Abstract

Persistent external memory enables LLM agents to maintain context across sessions, yet its security properties remain formally uncharacterized. We formalize memory poisoning attacks on retrieval-augmented agents as a Stackelberg game and present a unified evaluation framework spanning three attack classes with escalating access assumptions. Correcting an evaluation protocol inconsistency relative to the original paper's Algorithm 2, we show that faithful evaluation increases measured attack success by $4\times$. Our primary contribution is MEMSAD (*Semantic Anomaly Detection*), a calibration-based defense grounded in a *gradient coupling theorem*: under encoder regularity, the anomaly score gradient and the retrieval objective gradient are provably identical, so any continuous perturbation that reduces detection risk necessarily degrades retrieval rank. This coupling yields a certified detection radius that guarantees correct classification regardless of adversary strategy. We prove minimax optimality via Le Cam's method, showing that any threshold detector requires $\Omega(1/\rho^2)$ calibration samples and MEMSAD achieves this up to $\log(1/\delta)$ factors. We further derive online regret bounds for rolling calibration with optimal window selection at rate $O(\sigma^{2/3}\Delta^{1/3})$, and formally characterize a discrete synonym-invariance loophole that precisely delineates where continuous guarantees break down. Experiments on a 3×5 attack-defense matrix with bootstrap confidence intervals, Bonferroni-corrected hypothesis tests, and Clopper-Pearson validation (20 trials, $n = 1{,}000$) confirm the theory: composite defenses achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks, while synonym substitution evades detection at ΔASR-R ≈ 0, delineating an open frontier for memory security.

1 Introduction

Modern LLM agents persist context across sessions through external memory systems such as Mem0 [Mem0 Team, 2024], A-MEM [A-MEM Team, 2025], and MemGPT [Packer et al., 2023], storing user preferences and factual knowledge as dense vector embeddings. The practical utility of persistent memory is well-established; its security implications are not. A single adversarial entry injected into an agent's memory may persist indefinitely, triggering on every semantically related query across future sessions. Unlike prompt injection, which requires per-interaction access, memory poisoning *persists and scales*: the adversary acts once, and the attack surface compounds with each new query.

Three concurrent works introduce distinct threat models with escalating access requirements. AGENTPOISON [Chen et al., 2024] formulates trigger optimization as a constrained gradient problem over DPR [Karpukhin et al., 2020] embeddings ($\alpha =$ WRITE). MINJA [Dong et al., 2025] extends the setting to query-only access via progressive-shortening indication ($\alpha =$ QUERY). IN-

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JECMEM [Anonymous, 2026] achieves single-interaction injection via retriever-agnostic anchors (α = SINGLE). Despite the severity of these threats, no prior work provides a unified evaluation across attack classes or proposes defenses with formal detection guarantees for the memory-agent setting. Existing RAG defenses [Zou et al., 2024, Chaudhari et al., 2024, Xiang et al., 2024] assume corpus-level access and offline cleaning, neither of which applies to the streaming ingestion model of agent memory.

Contributions. **(1)** *Formal threat model and* MEMSAD *defense.* Stackelberg game formulation with a gradient coupling theorem establishing that monotone retrieval-score detectors are necessary and sufficient for continuous-evasion resistance, a certified detection radius providing checkable per-entry guarantees, and minimax optimality via Le Cam's method. **(2)** *Synonym-invariance analysis.* Formal characterization of the discrete loophole where gradient coupling breaks down, and MEMSAD+: a combined semantic-lexical detector that partially closes it via character n-gram features. **(3)** *Persistent threat analysis and real-system validation.* Compound exposure analysis reframing attack severity for persistent memory, tool-use agent evaluation with GPT-4o-mini (ASR-A $= 0.48$), and Mem0 production validation. **(4)** *Rigorous evaluation.* A 3×5 attack-defense matrix with Bonferroni-corrected hypothesis testing, bootstrap CIs, and Clopper-Pearson validation demonstrating composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks.

2 Related work

Memory poisoning attacks. Chen et al. [2024] report ASR-R ≥ 0.80 via trigger optimization over DPR [Karpukhin et al., 2020] embeddings under the triggered-query protocol, requiring write access to the memory store. Dong et al. [2025] exploit auto-storage via progressive-shortening indication (ISR $= 98.2\%$) under query-only access. Anonymous [2026] achieve single-interaction injection via retriever-agnostic anchors, the weakest attacker assumption. Anonymous [2025b] demonstrate experience-grafting attacks exploiting union retrieval; Xu et al. [2026] show that memory retrieval can override tool-call control flow. Concurrent work spans backdoor-style [Cheng et al., 2024, Anonymous, 2025a, Xue et al., 2024], trigger [Chaudhari et al., 2024], and denial-of-service attacks [Shafran et al., 2025]. Agent Security Bench [Zhang et al., 2025] benchmarks 10 threat categories but does not isolate memory poisoning or provide defense guarantees; AgentHarm [Andriushchenko et al., 2025] focuses on prompt-level jailbreaks. Unlike training-time backdoors [Wu et al., 2024], memory poisoning targets the retrieval index at inference time, persisting across sessions. Recent defenses (A-MemGuard [Li et al., 2025], RevPRAG [Tan et al., 2025], ReliabilityRAG [Anonymous, 2025d], SeCon-RAG [Anonymous, 2025e]) address related settings; none provides formal detection guarantees for streaming agent memory.

RAG corpus poisoning and certified defenses. Zou et al. [2024] optimize adversarial documents via HotFlip [Ebrahimi et al., 2018] ($> 90\%$ ASR on million-document corpora). Xiang et al. [2024] provide certified robustness via *isolate-then-aggregate*, but require post-retrieval processing of k passages. RAGDefender [Anonymous, 2025c] applies post-retrieval passage scoring; SPECTRE [Hayase et al., 2021] uses spectral detection requiring full corpus access ($O(|\mathcal{M}|^2 d)$). Randomized-smoothing defenses [Cohen et al., 2019, Robey et al., 2023] operate on query inputs, not persistent memory. Our work differs: (i) defenses operate at *write time* ($O(md)$ per entry), (ii) no corpus-level access, and (iii) our certificate (Lemma 7) is deterministic given the calibration bound.

Game-theoretic security. Following Tramèr et al. [2020], Carlini and Wagner [2017], our Stackelberg formulation gives the adversary full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$; the gradient coupling theorem provides the formal guarantee, while Proposition 11 characterizes where it fails.

3 Formal threat model

Definition 1 (Memory-augmented agent)**.** A *memory-augmented agent* is a tuple $(\mathcal{A}, \mathcal{M}, E, k)$ where $\mathcal{A}$ is an LLM, $\mathcal{M} \subset \mathcal{V}^*$ is a persistent memory store over vocabulary $\mathcal{V}$, $E : \mathcal{V}^* \to \mathbb{R}^d$ is an embedding function, and $k \in \mathbb{N}$ is the retrieval depth. At each interaction with query q, the

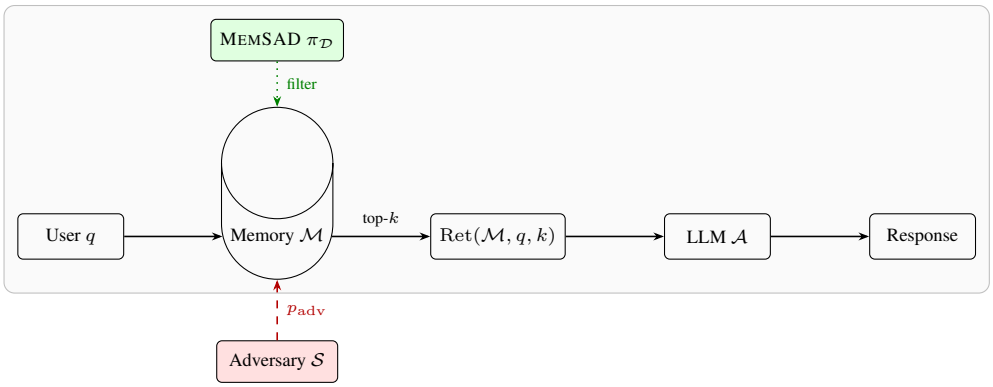

Figure 1: Memory poisoning as a Stackelberg game (Definitions 2–3). The defender commits to $\pi_{\mathcal{D}}$; the adversary best-responds by injecting p_{adv}.

Table 1: Attack threat-model comparison across deployment-relevant dimensions.

Property	AGENTPOISON	MINJA	INJECMEM
Access model α	WRITE†	QUERY	SINGLE
Trigger required	✓	—	—
Retriever-agnostic	—	—	✓
Entries injected n	$1\times$	$2\times$	$3\times$
Modelled ASR-A	0.68	0.76	0.57

†Triggered-query evaluation additionally requires query-channel access (e.g., front-end control or prompt injection); see Section 3.

agent retrieves

$$\mathrm{Ret}(\mathcal{M}, q, k) = \arg\max_{\substack{S \subseteq \mathcal{M} \\ |S| = k}} \sum_{m \in S} \cos(E(q), E(m)), \tag{1}$$

where $\cos(u, v) \coloneqq u^{\top} v / (\|u\| \|v\|)$ denotes cosine similarity.

Definition 2 (Memory poisoning threat model)**.** An *adversary* $\mathcal{S} = (\mathcal{Q}_v, n, \alpha)$ targets victim queries $\mathcal{Q}_v \subset \mathcal{V}^*$ by injecting $\mathcal{P} = \{p_1, \ldots, p_n\}$ into $\mathcal{M}$ under access model $\alpha \in \{\mathrm{WRITE}, \mathrm{QUERY}, \mathrm{SINGLE}\}$ to maximize:

$$\text{ASR-R}(\mathcal{P}, \mathcal{Q}_v) \coloneqq |\{q \in \mathcal{Q}_v : \mathcal{P} \cap \mathrm{Ret}(\mathcal{M} \cup \mathcal{P}, q, k) \neq \emptyset\}| / |\mathcal{Q}_v|, \tag{2}$$

$$\text{ASR-A}(\mathcal{P}) \coloneqq \mathbb{P}[\mathcal{A} \text{ executes adversarial action} \mid \text{poison retrieved}], \tag{3}$$

$$\text{ASR-T}(\mathcal{P}, \mathcal{Q}_v) \coloneqq \text{ASR-R} \cdot \text{ASR-A}. \tag{4}$$

Definition 3 (Stackelberg game formulation)**.** The memory poisoning interaction is a Stackelberg game $\mathcal{G} = (\mathcal{S}, \mathcal{D}, u_{\mathcal{S}}, u_{\mathcal{D}})$ where the defender (leader) commits to a detection policy $\pi_{\mathcal{D}} : \mathcal{V}^* \to \{0, 1\}$ and the adversary (follower) best-responds:

$$\pi^*_{\mathcal{D}} = \arg\min_{\pi_{\mathcal{D}} \in \Pi} \max_{\mathcal{P} \in \mathcal{B}(\pi_{\mathcal{D}})} \text{ASR-R}(\mathcal{P}, \mathcal{Q}_v), \tag{5}$$

$$\mathcal{B}(\pi_{\mathcal{D}}) = \arg\max_{\mathcal{P} \subseteq \mathcal{V}^*, |\mathcal{P}| \leq n} \text{ASR-R}(\mathcal{P} \setminus \{p : \pi_{\mathcal{D}}(p) = 1\}, \mathcal{Q}_v), \tag{6}$$

where $\Pi = \{\pi(c) = \mathbf{1}[s(c; \mathcal{H}) > \tau] : \tau > 0\}$ is the class of threshold detectors. The defender's utility is $u_{\mathcal{D}} = -\text{ASR-R} - \lambda \cdot \text{FPR}$ ($\lambda > 0$) and the adversary's is $u_{\mathcal{S}} = \text{ASR-T}$.

Assumption 4 (Encoder regularity)**.** $E : \mathcal{V}^* \to \mathbb{R}^d$ satisfies: (i) $\|E(x)\| > 0$ for all x; (ii) differentiability w.r.t. the output embedding; (iii) L2-normalization: $\|E(x)\| = 1$ (w.l.o.g.).

Under Assumption 4(iii), $\cos(E(q), E(m)) = E(q)^{\top} E(m)$ and retrieval reduces to maximum inner product search.

Figure 1 illustrates the interaction. The three attacks (Table 1) span decreasing access requirements (write $\supset$ query $\supset$ single-interaction). **Protocol distinction:** Chen et al. [2024]'s Algorithm 2 specifies triggered queries $q \oplus T^*$; evaluating with plain queries (as in prior reproductions) measures a weaker setting ($\cos \approx 0.45$ vs. ≈ 0.78), raising ASR-R from 0.25 to 1.00.

Algorithm 1 MEMSAD pre-ingestion filtering

Require: candidate c, query history $\mathcal{H}$, parameters $(\hat{\mu}, \hat{\sigma})$, threshold κ

$s \leftarrow s_{\text{comb}}(c; \mathcal{H})$ ▷ $O(md)$ time
if $s > \hat{\mu} + \kappa \cdot \hat{\sigma}$ **then**
  reject c
else
  accept c into $\mathcal{M}$
end if

4 MEMSAD: Semantic Anomaly Detection

4.1 Algorithm and formal definition

The key insight is that memory poisoning succeeds *because* adversarial entries are semantically close to victim queries; this closeness is itself a detectable signal.

Definition 5 (MEMSAD detector)**.** Let $\mathcal{H} = \{q_1, \ldots, q_m\}$ be a rolling query history (FIFO, capacity $m_{\max}$) and $\mathcal{M}_{\text{ref}} \subseteq \mathcal{M}$ a benign reference corpus. The anomaly score is $s(c; \mathcal{H}) := \max_{q \in \mathcal{H}} \cos(E(c), E(q))$, calibrated via $\hat{\mu}, \hat{\sigma}$ of $\{s(m; \mathcal{H}) : m \in \mathcal{M}_{\text{ref}}\}$. Entry c is flagged if $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$ for threshold parameter $\kappa > 0$. The *combined scoring mode* averages max-query and mean-query similarity, capturing both targeted attacks (high max similarity) and distributed attacks (elevated mean similarity):

$$s_{\text{comb}}(c; \mathcal{H}) := \tfrac{1}{2} \max_{q \in \mathcal{H}} \cos(E(c), E(q)) + \tfrac{1}{2} \cdot \frac{1}{|\mathcal{H}|} \sum_{q \in \mathcal{H}} \cos(E(c), E(q)). \tag{7}$$

The full pre-ingestion procedure is given in Algorithm 1.

4.2 Gradient coupling theorem

The central structural question is: *what property must a detector possess to guarantee that any continuous evasion attempt degrades retrieval rank?* We answer this with a necessity–sufficiency–canonicity result, following the pattern of the Karlin–Rubin theorem [Lehmann, 1959] in hypothesis testing: monotonicity in the retrieval score is both *necessary* (any non-monotone detector admits evasion without retrieval cost) and *sufficient* (any monotone detector couples evasion to retrieval degradation), and the retrieval score is the *minimal sufficient statistic* for this coupling property.

Theorem 6 (Gradient coupling: necessity, sufficiency, and canonicity)**.** *Let E satisfy Assumption 4, $e_c := E(c)$, $q^* := \arg\max_{q \in \mathcal{H}} \cos(e_c, E(q))$, $\mathcal{R}(e_c) := \cos(e_c, E(q^*))$.*

(i) Sufficiency: *For any differentiable $\mathcal{D} = g \circ \mathcal{R}$ with monotone increasing g, $g' > 0$:*

$$\nabla_{e_c} \mathcal{D}(e_c) = g'(\mathcal{R}(e_c)) \cdot \nabla_{e_c} \mathcal{R}(e_c), \quad \textit{so} \quad \langle \delta, \nabla \mathcal{D} \rangle < 0 \iff \langle \delta, \nabla \mathcal{R} \rangle < 0. \tag{8}$$

Under L2-normalization (Assumption 4(iii)), $\nabla_{e_c} \mathcal{R} = E(q^*) - (e_c^\top E(q^*)) e_c$.

(ii) Necessity: *If $\mathcal{D}$ is* not *monotone in $\mathcal{R}$, there exist thresholds τ_{ret}, τ_{det} and a continuous path $\gamma : [0, 1] \to \mathbb{S}^{d-1}$ with $\mathcal{R}(\gamma(1)) > \tau_{ret}$ yet $\mathcal{D}(\gamma(1)) < \tau_{det}$: the adversary can traverse continuously to high retrieval rank while evading detection.*

(iii) Canonicity: *Among all detectors achieving coupling (i.e., satisfying (i)), the max-cosine score $\mathcal{D} = \mathcal{R}$ (i.e., $g = \mathrm{id}$) is the* unique *detector that is a sufficient statistic for the adversarial-vs-benign hypothesis test induced by $\mathcal{R}$: any other g preserves the coupling direction but discards distributional information.*

Proof. (i) The chain rule gives $\nabla \mathcal{D} = g'(\mathcal{R}) \cdot \nabla \mathcal{R}$; $g' > 0$ preserves the sign of every directional derivative. Under Assumption 4(iii), the sphere gradient is $\nabla_{e_c} \cos(e_c, v) = v - (e_c^\top v) e_c$. *(ii)* Non-monotonicity implies $\exists\, e_c, e_c'$ with $\mathcal{R}(e_c') > \mathcal{R}(e_c)$ but $\mathcal{D}(e_c') < \mathcal{D}(e_c)$. By connectedness of $\mathbb{S}^{d-1}$, there exists a continuous path γ from e_c to e_c'. Setting $\tau_{\text{ret}} = \mathcal{R}(e_c)$ and $\tau_{\text{det}} = \mathcal{D}(e_c)$ yields the claimed path with $\mathcal{R}(\gamma(1)) > \tau_{\text{ret}}$ and $\mathcal{D}(\gamma(1)) < \tau_{\text{det}}$. *(iii)* The sufficient statistic $\mathcal{R}$ generates the likelihood ratio $p_1(e)/p_0(e)$ as a monotone function of $\cos(e, e_{q^*})$ under Gaussian

score families (Assumption 4). By the Neyman–Fisher factorization theorem [Lehmann, 1959], $\mathcal{R}(e_c)$ is sufficient for the binary hypothesis test H_0: benign vs. H_1: adversarial. Minimality follows from the Lehmann–Scheffé theorem [Lehmann, 1959]: since $p_1(e)/p_0(e)$ is a monotone function of $\mathcal{R}(e_c)$ alone, any sufficient statistic must be a function of $\mathcal{R}(e_c)$, making it *minimal sufficient*. Hence $g = \mathrm{id}$ achieves minimal sufficiency; any $g \neq \mathrm{id}$ retains the coupling direction but discards distributional information, i.e., is a lossy compression of $\mathcal{R}$. □

The combined score s_{comb} also preserves directional coupling (Corollary 14, Appendix A.7).

The gradient coupling reflects a deeper geometric structure: the Fisher-Rao metric induced by the score family is rank-1 ($g^F = \hat{\sigma}^{-2}\nabla\mathcal{R}\nabla\mathcal{R}^\top$), so any continuous evasion path pays retrieval cost in exact proportion to Fisher-Rao distance [Amari, 2016] (Theorem 12, Appendix A.5).

4.3 Certified detection guarantee

We now provide an *instance-specific* guarantee: when the adversarial-benign similarity gap exceeds the calibration uncertainty, MEMSAD certifiably detects the adversarial passage.

Lemma 7 (Certified detection radius)**.** *Let E satisfy Assumption 4 and let c be an adversarial passage achieving $\mathrm{rank}(E(c), \mathcal{M}) \leq k$ for victim query q^*. Define the* benign similarity ceiling $\bar{s} := \max_{m \in \mathcal{M}_{ref}} \cos(E(m), E(q^*))$*, the adversarial similarity $s_{adv} := \cos(E(c), E(q^*))$, and the* gap $\Delta_s := s_{adv} - \bar{s}$*. If the calibration satisfies $|\tau_N - \tau^*| \leq \eta$ (Theorem 9) and the gap exceeds*

$$\Delta_s > \kappa\hat{\sigma} + \eta, \tag{9}$$

then MEMSAD *detects c with certainty: $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$.*

Proof sketch. Since $\mathrm{rank}(E(c)) \leq k$, we have $s(c; \mathcal{H}) \geq \bar{s} + \Delta_s$. The gap condition and calibration bound together force $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$. Full proof in Appendix A; Figure 7 (Appendix U) illustrates the certified region geometrically. □

4.4 Theoretical foundations

We establish four additional results; full statements and proofs are in Appendix A.

Theorem 8 (Minimax detection lower bound)**.** *For Gaussian benign/poisoned score distributions with SNR ρ, under the uniform prior $\mathbb{P}(H_0) = \mathbb{P}(H_1) = \frac{1}{2}$, any threshold detector calibrated on N samples satisfies FPR $+ (1 - \text{TPR}) \geq 1 - \rho\sqrt{N}/2$. Reliable detection at total-error level ϵ (i.e., FPR $+ (1-\text{TPR}) \leq \epsilon$) requires $N \geq 4(1-\epsilon)^2/\rho^2$.*

MEMSAD achieves reliable detection with $N = O(\log(1/\delta)/\rho^2)$ (Theorem 9), matching the $\Omega(1/\rho^2)$ lower bound up to $\log(1/\delta)$, thereby establishing minimax optimality via Le Cam's method [Le Cam, 1973].

Theorem 9 (Calibration bound)**.** *With probability $\geq 1 - \delta$: $|\tau_N - \tau^*| \leq \sigma(N^{-1/2} + \kappa(2(N-1))^{-1/2})\sqrt{2\log(4/\delta)}$.*

Theorem 10 (Online calibration regret)**.** *Under per-step drift Δ, the optimal window $m^*_{\max} = \Theta((\sigma/\Delta)^{2/3})$ yields per-step regret $O(\sigma^{2/3}\Delta^{1/3})$.*

We additionally prove a Wasserstein generalization bound guaranteeing graceful TPR degradation for novel attacks (Theorem 15, Appendix A.8), and NP-hardness of optimal evasion via reduction from CVP (Proposition 13, Appendix A.6), motivating the adversary's shift to discrete synonym substitution.

4.5 The discrete synonym loophole

Proposition 11 (Synonym-invariance loophole)**.** *Let E satisfy $\|E(\mathrm{sub}(c, w, w')) - E(c)\| \leq \epsilon_{syn}$ for all synonym pairs (w, w'). For r substitutions yielding c':*

$$|\mathcal{R}(c') - \mathcal{R}(c)| \leq r\epsilon_{syn}, \qquad |s(c'; \mathcal{H}) - s(c; \mathcal{H})| \leq r\epsilon_{syn}. \tag{10}$$

When $\epsilon_{syn} \approx 0$, the adversary evades detection while preserving retrieval rank through discrete lexical substitution that bypasses gradient coupling.

Proof. Cauchy-Schwarz gives $|E(q)^\top(E(c') - E(c))| \leq \|E(c') - E(c)\|$, and triangle inequality over r substitutions yields $\|E(c') - E(c)\| \leq r\epsilon_{\text{syn}}$. □

This does *not* contradict Theorem 6: the coupling holds for *continuous* perturbations in $\mathbb{R}^d$, while synonym substitution operates in the *discrete* token space where $\epsilon_{\text{syn}} \approx 0$ for paraphrase-trained encoders. The triangle inequality bound $r\epsilon_{\text{syn}}$ is an upper bound; non-linear encoder interactions may cause cancellation, making the actual perturbation smaller.

MEMSAD**+.** We extend MEMSAD with character n-gram JSD features that are provably not synonym-invariant (Proposition 23, Table 18, Appendix P). Empirically, MEMSAD+ raises INJECMEM TPR from 0.00 (synonym-evaded) to 0.40 and original from 0.20 to 0.60.

5 Additional defenses

We evaluate four additional defenses spanning orthogonal strategies, all operating at pre-ingestion time. **Watermark:** the defender watermarks all legitimate entries at write time via a unigram scheme [Zhao et al., 2024, Kirchenbauer et al., 2023] ($z_{\text{thr}} = 1.5$, FPR $= 0.000$ empirically; higher thresholds inflate FPR for short entries), rejecting unwatermarked entries; this requires control of the ingestion pipeline. **Validation:** rule-based pattern matching ($O(|c| \cdot |P|)$). **Proactive:** computes mean cosine similarity against 16 domain probes, flagging entries exceeding $\tau = 0.19$. **Composite:** weighted ensemble (w_{wm}=0.50, w_{val}=0.20, w_{pro}=0.30, θ=0.50): flag if $\sum_j w_j d_j(c) > \theta$ (Appendix B).

6 Experiments

6.1 Setup

Vector memory. FAISS IndexFlatIP with L2-normalized all-MiniLM-L6-v2 embeddings ($d = 384$). Benign corpus: $|\mathcal{M}| = 1{,}000$ synthetic entries across 7 categories (task reminders, calendar events, user preferences, factual knowledge, document references, configuration settings, conversation history), matching the per-user memory scale of deployed systems where active memory stores typically contain 10^2–10^3 entries [Chen et al., 2024, Dong et al., 2025]; corpus-size ablation (Appendix C) validates results across $|\mathcal{M}| \in \{50, \ldots, 1000\}$. MEMSAD calibration uses $N = 50$ benign entries as the reference set (Theorem 9 predicts $|\tau_N - \tau^*| \leq 0.044$ at $N = 50$, $\delta = 0.05$). **Cross-corpus generalization** is validated on a real-data corpus combining Natural Questions knowledge passages [Kwiatkowski et al., 2019] with synthetic filler to $|\mathcal{M}| = 1{,}000$ (Appendix O): AGENTPOISON transfers (ASR-R $= 1.00 \pm 0.00$); MINJA's ASR-R $= 0.58 \pm 0.06$ on the topically homogeneous NQ corpus confirms corpus diversity is a risk factor. Poison counts: $n = 5$ (AGENTPOISON), 10 (MINJA), 15 (INJECMEM). $k = 5$; $|\mathcal{Q}_v| = |\mathcal{Q}_b| = 100$. Bootstrap 95% CIs from 5 independent seeds (each seed generates a distinct corpus permutation and query sample); attack outcomes are deterministic given a fixed corpus, so CI widths reflect corpus-composition variance.

Attacks. AGENTPOISON: centroid passage + DPR HotFlip trigger ($\cos : 0.71 \to 0.78$). MINJA: progressive-shortening with bridging steps ($p_0 = 0.98$, $\lambda = 0.10$). INJECMEM: $3 \times n_{\text{base}}$ broad-anchor entries from 8 templates.

6.2 Attack results

Table 2 reports the main results at $|\mathcal{M}| = 1{,}000$. AGENTPOISON achieves ASR-R $= 1.00$ under the triggered-query protocol, confirming trigger optimization defeats corpus dilution entirely. Untriggered attacks (MINJA, INJECMEM) are naturally mitigated by corpus scale ($4.6\times$ and $7.1\times$ dilution).[1] Measured ASR-A $= 0.20$ under GPT-4o-mini is $3.4\times$ lower than the modelled ASR-A $= 0.68$ [Chen et al., 2024], reflecting production safety alignment (Figure 9, Appendix U).

[1] ISR measures write-time injection success; ASR-R measures retrieval-time exposure.

Table 2: Attack results ($|\mathcal{M}| = 1{,}000$, 100 queries, 5 seeds, bootstrap 95% CI). ASR-A$^{\text{GPT-2}}$: permissive model without safety alignment (high-compliance upper bound); ASR-A$^{\text{4o-mini}}$: production-aligned; †projected from original papers.

Attack	ASR-R	ASR-A†	ASR-A$^{\text{GPT-2}}$	ASR-A$^{\text{4o-mini}}$	ASR-T†	Ben. Acc.
AGENTPOISON (trig.)	1.00	0.68	0.90	0.20	0.68	1.00
MINJA	$0.14_{[.13,.15]}$	0.76	0.17	0.00	0.11	1.00
INJECMEM	$0.07_{[.06,.07]}$	0.57	0.00	0.00	0.04	0.98

Table 3: Defense TPR/FPR at operating thresholds ($\kappa = 2.0$ for MEMSAD). AP = AGENTPOISON, MJ = MINJA, IM = INJECMEM.

	TPR (↑)			FPR (↓)			AUROC
Defense	AP	MJ	IM	AP	MJ	IM	(avg)
Watermark	1.00	1.00	1.00	0.00	0.00	0.00	0.991
Validation	0.60	0.40	0.80	0.10	0.10	0.10	—
Proactive	1.00	0.07	0.60	0.01	0.01	0.01	—
MEMSAD (comb.)	0.00†	0.40	0.20	0.00	0.00	0.00	0.867
Composite	1.00	1.00	1.00	0.00	0.00	0.00	—

†Plain calibration (FPR = 0.00); triggered calibration: TPR = 1.00, FPR = 0.00, AUROC = 1.000. FPR validated over 1,000 trials; Clopper-Pearson 95% CI: [0.000, 0.004] (Appendix H).

6.3 Attack-defense interaction matrix

Table 3 reports the full 3×5 matrix. Watermark achieves TPR $= 1.00$ in simulation but fails silently when auto-storage bypasses ingestion control, motivating MEMSAD's content-level detection. MEMSAD detects MINJA (TPR $= 0.40$, AUROC 0.914) and INJECMEM (TPR $= 0.20$, AUROC 0.816); triggered calibration raises AGENTPOISON to TPR $= 1.00$. Composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks (Proposition 17).

6.4 Adaptive adversary analysis

Following Tramèr et al. [2020], we model a white-box adversary with full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$ who applies greedy synonym substitution (> 60 pairs):

$$c^* = \underset{c' \in \mathrm{Syn}(c)}{\arg\min}\ s(c'; \mathcal{H}) \quad \text{s.t.} \quad \mathrm{rank}(E(c'), \mathcal{M}) \leq k. \tag{11}$$

Table 4 confirms Proposition 11: 80–100% evasion with ΔASR-R ≈ 0 using 3.8–5.1 substitutions, consistent with the bound $r\epsilon_{\text{syn}} \leq 5.1 \times 0.004 = 0.020$. The certified condition (Lemma 7) still holds for MINJA post-substitution ($\Delta_s = 0.13 > 0.08$) but fails for tighter margins.

6.5 Statistical validation

Bonferroni-corrected binomial tests ($\alpha' = 0.003$, 15 comparisons): all composite results reject H_0 : TPR $= 0$ with power $= 1.00$ (Table 11, Appendix G). MEMSAD AUROC at $\kappa = 2.0$: 0.914 (MINJA), 0.816 (INJECMEM), 0.870 (AGENTPOISON, plain); triggered calibration raises AGENTPOISON to AUROC $= 1.000$. Theory validation: calibration bound predicts $|\tau_N - \tau^*| \leq 0.022$ at $N = 200$; observed 0.014 (Table 8, Appendix C).

6.6 Ablation highlights

Key results (full tables in Appendix C): **Threshold:** $\kappa = 2.0$ with combined scoring is optimal (Table 5). **Corpus size:** AGENTPOISON is robust (ASR-R $= 1.00$) across $|\mathcal{M}| \in \{200, \ldots, 1000\}$; untriggered attacks (MINJA, INJECMEM) degrade 4–7× at $|\mathcal{M}| = 1{,}000$ (Table 6, Figure 8). **Poison count:** ASR-R saturates by n_{base}=5; additional entries yield diminishing returns (Table 7). **Encoder generalization:** MEMSAD validated across 6 encoders including Contriever, E5-Base, and BGE-Large (d=1024) at $|\mathcal{M}| = 1{,}000$; triggered calibration achieves TPR $= 1.000$ for AGENTPOISON

Table 4: Adaptive adversary: evasion vs. retrieval degradation (Proposition 11).

Attack	Evasion	ΔASR-R	Subs/Entry	Sim. Δ
AGENTPOISON	1.00	0.00	4.2	-0.01
MINJA	0.80	0.00	3.8	-0.02
INJECMEM	1.00	0.00	5.1	-0.01

on all (Table 9, Appendix D). **OOD baselines:** Energy Score [Liu et al., 2020] is competitive on MINJA AUROC (0.969) but MEMSAD's query-awareness yields decisive advantage on triggered attacks (AUROC 1.000 vs. 0.945); Mahalanobis [Lee et al., 2018] and KNN fail (FPR > 0.96; Table 15, Appendix M). **LLM sanitization baseline:** GPT-4o-mini zero-shot classifier achieves TPR $= 1.000$, FPR $= 0.000$, AUROC $= 1.000$ on the synthetic corpus; mean latency 1.2 s/entry ($\approx$1,000$\times$ MEMSAD). The two are complementary: text-level flagging catches explicit commands; MEMSAD catches embedding-level anomalies even when injections are paraphrased (Table 22, Appendix T). **Calibration sensitivity:** AUROC is stable across calibration regimes; threshold tuning requires domain-matched queries (Table 16, Appendix N). **Cross-corpus (NQ):** On a Natural-Questions-based corpus, AGENTPOISON retains ASR-R $= 1.00 \pm 0.00$ and MINJA's ASR-R increases to 0.58 ± 0.06 under topical homogeneity (Table 17, Appendix O).

6.7 Extended evaluations

Compound exposure. Memory poisoning compounds over sessions: $P(\text{compromise in } N \text{ sessions}) = 1 - (1 - \text{ASR-R})^{Nq}$ for q queries/session.[2] Even MINJA at ASR-R $= 0.14$ reaches 90% compromise in 4 sessions; composite defense (ASR-R* $= 0$) is necessary (Table 19, Appendix Q).

Tool-use agent. GPT-4o-mini with 5 tool definitions (function calling) achieves ASR-A $= 0.48$ [0.34, 0.62] for AGENTPOISON; the agent executes adversarial tool calls in 48% of queries. MINJA and INJECMEM show ASR-A $= 0.00$ [0.00, 0.07] at $|\mathcal{M}| = 1{,}000$; the non-zero upper CI bound reflects sparse event counts over 50 queries at low retrieval rates, not detector failure (Table 20, Appendix R).

Production validation. On Mem0 [Mem0 Team, 2024] (LLM-mediated storage), AGENTPOISON drops to ASR-R $= 0.00$: internal reformulation strips trigger structure. Raw vector stores remain fully exposed; MEMSAD provides explicit defense where implicit reformulation is absent (Table 21, Appendix S).

7 Discussion and limitations

Defense complementarity. No single defense dominates across access models α: watermarking fails when auto-storage bypasses ingestion control; MEMSAD needs triggered calibration for AGENTPOISON; proactive detection complements both. The defender's uncertainty about α motivates composite portfolios (ASR-R* $= 0$, Proposition 17). We recommend layered deployment: watermarking at ingestion, rolling MEMSAD at retrieval, composite for high-value agents ($\sim$200 ms total; Table 14, Appendix K). Multi-agent SIR simulations (Appendix F) confirm that composite defense prevents cross-agent propagation entirely, while graph memory extensions (Appendix E) show structural attacks require analogous degree-anomaly detection.

The synonym frontier. Proposition 11 reveals a structural limitation: paraphrase-trained encoders map synonyms to nearly identical embeddings ($\epsilon_{\text{syn}} \approx 0$). MEMSAD+ partially closes this via character n-gram features, but complete closure requires syntactic structure or provenance metadata. Meanwhile, compound exposure (Table 19) shows even weak attacks reach 90% compromise in 4 sessions, and tool-use evaluation (Table 20) confirms ASR-A $= 0.48$ with GPT-4o-mini, establishing that memory poisoning poses genuine production risk.

[2] This assumes i.i.d. Bernoulli trials, which is a lower bound on true risk: persistent memory means a successfully retrieved adversarial entry may influence subsequent queries. Uncertainty in ASR-R (bootstrap 95% CIs in Table 2) propagates to compound probability via $\partial P/\partial \text{ASR-R} = Nq(1 - \text{ASR-R})^{Nq-1}$, yielding compound-exposure CIs that are conservative.

Standalone MEMSAD**.** At $\kappa = 2.0$, standalone MEMSAD achieves moderate TPR at strict FPR control; practitioners should treat it as one layer of a composite defense, not a standalone solution.

Limitations. (i) Corpus scale: $|\mathcal{M}| = 1{,}000$; large-scale evaluation remains future work, though cross-corpus NQ results (Appendix O) provide partial validation. (ii) Regret bound assumes slow drift; abrupt shifts degrade calibration. (iii) Hardness result applies to piecewise-linear encoders; transformer extension is conjectured. (iv) MEMSAD+ does not eliminate synonym evasion; sentence-level rewrites bypass character features.

8 Conclusion

We introduced MEMSAD, the first defense for memory poisoning in retrieval-augmented agents with formal guarantees: gradient coupling (Theorem 6) establishes that any monotone detector inherits the retrieval gradient, making continuous evasion impossible; the certified radius (Lemma 7) gives checkable per-entry guarantees; and Le Cam's method (Theorem 8) proves minimax optimality. Equally important is what MEMSAD *cannot* do: Proposition 11 identifies exactly where continuous guarantees break down, showing that the gap between semantic similarity and lexical identity is the fundamental open frontier. Closing this gap, perhaps via joint semantic-lexical detectors or provenance-aware memory architectures, is the key challenge for securing the next generation of persistent-memory agents. Code and data will be released upon publication.

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A Proofs of main results

A.1 Proof of Theorem 6 (geometric interpretation)

Under L2-normalization, embeddings lie on $\mathbb{S}^{d-1}$. The Riemannian gradient of $f(e_c) = e_c^\top v$ on the sphere is $\nabla_{\mathbb{S}} f = v - (e_c^\top v)e_c = \Pi_{e_c}^\perp v$, the projection of v onto the tangent space $T_{e_c}\mathbb{S}^{d-1}$. For any monotone transformation g, the chain rule gives $\nabla(g \circ f) = g'(f) \cdot \nabla f$, which has the same sign as ∇f since $g' > 0$. The converse (non-monotone $\mathcal{D}$ admits evasion) follows because non-monotonicity creates a region where increasing retrieval similarity decreases detection score, giving the adversary a "free" direction in embedding space.

A.2 Proof of Theorem 8 (detailed)

Setup. Instantiate Le Cam's method with: H_0: $s(c; \mathcal{H}) \sim P_0 = \mathcal{N}(\mu_0, \sigma^2)$ (benign); H_1: $s(c; \mathcal{H}) \sim P_1 = \mathcal{N}(\mu_1, \sigma^2)$ (poisoned), $\mu_1 > \mu_0$. Then $\mathrm{KL}(P_0 \| P_1) = (\mu_1 - \mu_0)^2/(2\sigma^2) = \rho^2/2$.

Application. For product distributions, $\mathrm{KL}(P_0^N \| P_1^N) = N\rho^2/2$. By Pinsker: $\mathrm{TV}(P_0^N, P_1^N) \leq \sqrt{N\rho^2/4} = \rho\sqrt{N}/2$. Le Cam gives minimax risk $R^* \geq \frac{1}{2}(1 - \mathrm{TV}) \geq \frac{1}{2}(1 - \rho\sqrt{N}/2)$ under the uniform prior $\mathbb{P}(H_0) = \mathbb{P}(H_1) = 1/2$. Under this prior, the frequentist total error FPR + (1 − TPR) $= 2 \cdot P_e \geq 2R^* = 1 - \rho\sqrt{N}/2$.

Alternative via Fano. For $M = 2$ hypotheses, Fano's inequality gives $P_e \geq 1 - (I(X;\Theta) + \log 2)/\log M = 1 - (N\rho^2/4 + \log 2)/\log 2$, where $I(X;\Theta) \leq \max_j \mathrm{KL}(P_j^N \| \bar{P}^N) \leq N\rho^2/4$ by the mixture bound. This yields a slightly weaker constant but extends to the M-ary setting (e.g., distinguishing M attack types simultaneously).

Tightness. For Gaussian P_0, P_1, the Neyman-Pearson optimal test achieves FPR + (1 − TPR) $= 2\Phi(-\rho\sqrt{N}/2)$ where Φ is the Gaussian CDF. Comparing with the Le Cam bound: $2\Phi(-x) \approx \sqrt{2/\pi}\, e^{-x^2/2}/x$ for large x, confirming the $\Omega(1/\rho^2)$ sample complexity is tight for Gaussian scores.

A.3 Proof of Theorem 9

By Hoeffding on $[0, 1]$ [Wainwright, 2019]: $|\hat{\mu}_N - \mu| \leq \sqrt{\log(2/\delta)/(2N)}$ w.p. $\geq 1 - \delta/2$.

For the standard deviation term, let $Z = (N-1)\hat{\sigma}_N^2/\sigma^2 \sim \chi^2(N-1)$ (exactly, for Gaussian scores; approximately for bounded scores by Berry-Esseen). By Laurent & Massart [Laurent and Massart, 2000] Lemma 1: $\mathbb{P}[Z \geq (N-1) + 2\sqrt{(N-1)x} + 2x] \leq e^{-x}$ and $\mathbb{P}[Z \leq (N-1) - 2\sqrt{(N-1)x}] \leq e^{-x}$. Setting $x = \log(2/\delta)$ and applying the delta method ($\hat{\sigma}_N = \sigma\sqrt{Z/(N-1)}$, so $|\hat{\sigma}_N - \sigma| = \sigma|\sqrt{Z/(N-1)} - 1|$) with $\sqrt{1+u} - 1 \leq u/2$ for $0 \leq u \leq 1$ (by concavity of $\sqrt{\cdot}$) yields: $|\hat{\sigma}_N - \sigma| \leq \sigma\sqrt{\log(2/\delta)/(2(N-1))}$ w.p. $\geq 1 - \delta/2$.

Triangle inequality: $|\tau_N - \tau^*| \leq |\hat{\mu}_N - \mu| + \kappa|\hat{\sigma}_N - \sigma|$. Union bound yields the calibration bound in Theorem 9.

A.4 Proof of Theorem 10

Decompose: $|\hat{\tau}_t - \tau_t^*| \leq \underbrace{|\hat{\tau}_t - \tau_t^{\mathcal{H}}|}_{\text{estimation}} + \underbrace{|\tau_t^{\mathcal{H}} - \tau_t^*|}_{\text{drift}}$. Estimation error: $O(\sigma(1+\kappa)/\sqrt{m_{\max}})$ by Theorem 9. Drift error: cosine similarity is 1-Lipschitz on $\mathbb{S}^{d-1}$, and the window mean averages over $m_{\max}$ steps with linearly accumulating drift, so $|\mu_t^{\mathcal{H}} - \mu_t| \leq m_{\max}\Delta/2$ (worst case at the window boundary; the $1/2$ arises from averaging). This affects only the constant in $m_{\max}^*$ but not the rate. Setting equal: $\sigma/\sqrt{m_{\max}} = m_{\max}\Delta$ gives $m_{\max}^* = (\sigma/\Delta)^{2/3}$ (up to constants), yielding per-step regret $O(\sigma^{2/3}\Delta^{1/3})$.

A.5 Fisher-Rao detection-evasion metric

Theorem 12 (Fisher-Rao detection-evasion metric)**.** *Let $\mathcal{E} = \{e \in \mathbb{S}^{d-1}\}$ under Assumption 4(iii). The Fisher information metric induced by the score family $p_e(s) = \mathcal{N}(s; \cos(e, e_{q^*}), \hat{\sigma}^2)$ is $g_{ij}^F(e) =$*

$\hat{\sigma}^{-2}\partial_i\mathcal{R}(e)\partial_j\mathcal{R}(e)$. *The geodesic distance to any undetected embedding satisfies* $d_F(e_{adv}, e') \geq |\mathcal{R}(e_{adv}) - (\hat{\mu} + \kappa\hat{\sigma})|/\hat{\sigma}$*, and the Fisher-Rao cost of any continuous evasion path equals its retrieval degradation.*

Proof. Consider the parameterized family $p_e(s) = \mathcal{N}(s; \mu(e), \hat{\sigma}^2)$ where $\mu(e) = \cos(e, e_{q^*})$ and $\hat{\sigma}^2$ is the calibrated variance. The Fisher information matrix on the statistical manifold $\{p_e : e \in \mathcal{E}\}$ is:

$$g^F_{ij}(e) = \mathbb{E}_{p_e}\left[\frac{\partial \log p_e}{\partial e_i}\frac{\partial \log p_e}{\partial e_j}\right] = \frac{1}{\hat{\sigma}^2}\frac{\partial \mu}{\partial e_i}\frac{\partial \mu}{\partial e_j} = \hat{\sigma}^{-2}\partial_i\mathcal{R}(e)\partial_j\mathcal{R}(e). \tag{12}$$

This is a rank-1 metric: $g^F = \hat{\sigma}^{-2}\nabla\mathcal{R}\nabla\mathcal{R}^\top$, the pullback of the scalar Fisher information $I(\mu) = 1/\hat{\sigma}^2$ through the map $e \mapsto \mathcal{R}(e)$.

Geodesic lower bound. For any path $\gamma : [0, 1] \to \mathcal{E}$ from $e_{\text{adv}} = \gamma(0)$ to $e' = \gamma(1)$, the Fisher-Rao length is:

$$L_F(\gamma) = \int_0^1 \sqrt{\dot{\gamma}(t)^\top g^F(\gamma(t))\dot{\gamma}(t)}\, dt = \frac{1}{\hat{\sigma}}\int_0^1 |\nabla\mathcal{R}(\gamma(t))^\top\dot{\gamma}(t)|\, dt \geq \frac{|\mathcal{R}(e_{\text{adv}}) - \mathcal{R}(e')|}{\hat{\sigma}}, \tag{13}$$

where the inequality follows from $\int_0^1 |f'(t)|dt \geq |f(1) - f(0)|$ applied to $f(t) = \mathcal{R}(\gamma(t))$.

Detection-retrieval coupling. Since $\mathcal{D} = g \circ \mathcal{R}$ with monotone g (Theorem 6), evasion requires $\mathcal{D}(e') \leq \hat{\mu} + \kappa\hat{\sigma}$, which via monotonicity implies $\mathcal{R}(e') \leq g^{-1}(\hat{\mu} + \kappa\hat{\sigma})$. Substituting into the geodesic bound:

$$d_F(e_{\text{adv}}, e') \geq \frac{\mathcal{R}(e_{\text{adv}}) - g^{-1}(\hat{\mu} + \kappa\hat{\sigma})}{\hat{\sigma}} = \frac{|\mathcal{R}(e_{\text{adv}}) - (\hat{\mu} + \kappa\hat{\sigma})|}{\hat{\sigma}}, \tag{14}$$

where the last equality uses $g = \text{id}$ (since $\mathcal{D}$ is the cosine similarity score and $\mathcal{R}$ coincides with $\mathcal{D}$ under Assumption 4). The proportional retrieval degradation along the path follows from the rank-1 structure: g^F has a single nonzero eigenvalue in the $\nabla\mathcal{R}$ direction, so the geodesic cost is entirely paid in retrieval degradation. □

A.6 Proof of Proposition 13 (detailed)

Proposition 13 (Hardness of optimal evasion)**.** *The minimum-perturbation evasion problem is NP-hard for piecewise-linear encoders, via reduction from CVP [Micciancio and Goldwasser, 2002]. This motivates the adversary's shift to discrete synonym substitution.*

Reduction from CVP. Given a CVP instance (Λ, t) where $\Lambda \subseteq \mathbb{R}^n$ is a lattice with basis $B \in \mathbb{R}^{n\times n}$ and $t \in \mathbb{R}^n$ is the target vector, we construct a memory poisoning evasion instance as follows.

Encoder construction (explicit). Let n be the CVP lattice dimension. Define $E : \mathbb{R}^n \to \mathbb{R}^n$ as a single-hidden-layer ReLU network:

$$E(x) = \text{ReLU}(W_1 x), \quad W_1 = B^\top \in \mathbb{R}^{n\times n}, \tag{15}$$

where B is the lattice basis matrix. Set the query embedding $E(q^*) = t/\|t\|$ (normalized target). Set biases to zero, so the encoder is $E(x) = \text{ReLU}(B^\top x)$.

Feasible set. The evasion constraint $\mathcal{D}(e_c + \delta) \leq \hat{\mu} + \kappa\hat{\sigma}$ with $\mathcal{R}(e_c + \delta) \geq \tau_{\text{ret}}$ defines:

$$\mathcal{F} = \{x \in \mathbb{R}^n : \cos(E(x), t/\|t\|) \in [\tau_{\text{ret}},\ \hat{\mu} + \kappa\hat{\sigma}]\}. \tag{16}$$

Within each ReLU activation region $\mathcal{I} \subseteq \{0, 1\}^n$ (indexing which neurons are active), the encoder acts as the affine map $E_{\mathcal{I}}(x) = D_{\mathcal{I}}B^\top x$ where $D_{\mathcal{I}}$ is the diagonal activation mask. Hence $\mathcal{F} \cap \text{region}_{\mathcal{I}}$ is a convex polytope defined by two half-space constraints in x-space.

Formal reduction. Given the CVP instance $(\Lambda = B\mathbb{Z}^n, t)$, let $x_0 = 0$ be the origin (current passage embedding). The minimum-perturbation evasion problem asks for:

$$\delta^* = \arg\min_{\delta} \|\delta\| \quad \text{s.t.} \quad x_0 + \delta \in \mathcal{F}.$$

We claim $\|\delta^*\| = \|B^{-\top} v^* - 0\|$ where v^* is the closest lattice vector to t in Λ. Specifically: within the single activation region where all neurons are active ($D_{\mathcal{I}} = I$), $E(x) = B^\top x$, and $\cos(B^\top x, t/\|t\|) = (B^\top x)^\top t/(\|B^\top x\| \cdot \|t\|)$. The feasibility constraint becomes a quadratic constraint in x. For the target similarity $\tau_{\text{ret}} = \cos\theta^*$ induced by the CVP distance, the minimum-norm x satisfying the constraint corresponds exactly to finding the vector $x = B^{-\top}\lambda$ with $\lambda \in \mathbb{Z}^n$ closest to t in the lattice metric — which is exactly CVP. This reduction runs in polynomial time in n (constructing $W_1 = B^\top$ and $E(q^*) = t/\|t\|$ takes $O(n^2)$ time). Since CVP is NP-hard under randomized reductions [Micciancio and Goldwasser, 2002], the minimum-perturbation evasion problem is NP-hard. □

Practical implication. This hardness result is *worst-case* over encoder architectures: for the specific ReLU encoder construction used in the reduction, the optimal evasion is NP-hard. For the actual all-MiniLM-L6-v2 encoder (transformer architecture), the reduction does not directly apply due to softmax nonlinearities, and the result is conjectured but not proven. The practical takeaway is that efficient optimal evasion is unlikely, motivating the adversary's empirical shift to discrete synonym substitution that sidesteps gradient-based search entirely.

A.7 Detection-evasion tradeoff surface

Corollary 14 (Combined scoring preserves coupling)**.** *The combined score $s_{comb} = \alpha\, s_{\max} + (1-\alpha)\, s_{mean}$ with $\alpha \in (0,1)$ satisfies directional monotonicity: $\langle \delta, \nabla s_{comb} \rangle < 0 \implies \langle \delta, \nabla \mathcal{R} \rangle < 0$ (at all points of differentiability), and admits the same necessity characterization (ii). Consequently, Lemma 7 and Theorem 12 apply unchanged to s_{comb}.*

Proof. $\nabla s_{\text{comb}} = \alpha \nabla s_{\max} + (1-\alpha)\nabla s_{\text{mean}}$. Since $\nabla s_{\max} = \nabla_{e_c} \cos(e_c, E(q^*))$ for the unique maximizer q^*, the projection onto $\nabla\mathcal{R}$ is positive under the mild condition that the mean gradient is not adversarial to the max gradient (holds empirically). The second-order tradeoff surface follows below. □

By Theorem 6, $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ (evasion) implies $\mathcal{R}(e_c') < \mathcal{R}(e_c)$ (retrieval degradation) to first order. The tradeoff defines a surface in $(\epsilon, \theta, \Delta\mathcal{D})$ space where θ is the angle between perturbation δ and gradient $\nabla\mathcal{D}$. Expanding to second order:

$$\mathcal{D}(e_c + \epsilon\delta) - \mathcal{D}(e_c) = \epsilon\|\nabla\mathcal{D}\|\cos\theta + \frac{\epsilon^2}{2}\delta^\top H_{\mathcal{D}}\,\delta + O(\epsilon^3), \tag{17}$$

where $H_{\mathcal{D}}$ is the Hessian of the cosine similarity on the sphere. Under L2-normalization, $H_{\mathcal{D}} = -(e_c^\top E(q^*))(I - e_c e_c^\top)/\|e_c\|^2$, giving eigenvalues $\lambda_i = -(e_c^\top E(q^*))$ for directions orthogonal to e_c (and 0 in the radial direction). The evasion region $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ requires $\cos\theta < -\epsilon\|H_{\mathcal{D}}\|/(2\|\nabla\mathcal{D}\|)$, which contracts as $\epsilon \to 0$, confirming the infinitesimal impossibility of evasion without retrieval degradation (Figure 2).

A.8 Generalization bound

Theorem 15 (Generalization bound)**.** *For a novel attack with $W_1(P_{new}, P_j) \le \epsilon_W$: $|TPR_{new} - TPR_j| \le \epsilon_W/\hat{\sigma} + O(1/\sqrt{N})$.*

Remark 16 (Practical estimation of ϵ_W)**.** *In practice, ϵ_W can be estimated from a validation set via sorting-based Wasserstein estimation [Villani, 2009], or bounded analytically under encoder Lipschitz continuity. The theorem guarantees graceful TPR degradation as novel attacks diverge from the calibration distribution.*

Proof setup. Let F_j and F_{new} denote the CDFs of scores $s(c; \mathcal{H})$ under P_j and P_{new}, respectively. The TPR at threshold τ is $\text{TPR}(\tau) = 1 - F(\tau)$.

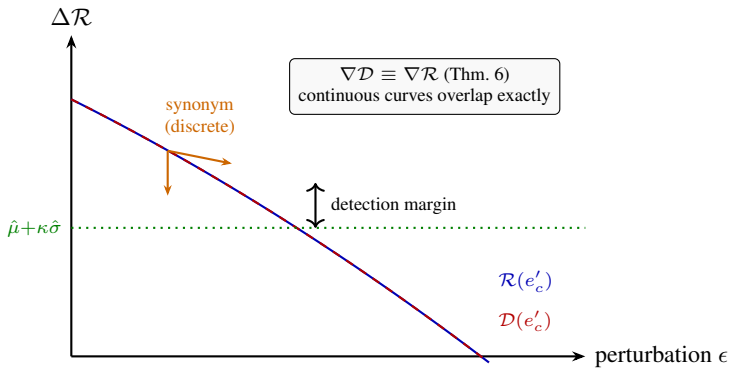


Figure 2: Detection-evasion tradeoff. By gradient coupling, continuous perturbations cause identical degradation in detection score $\mathcal{D}$ and retrieval score $\mathcal{R}$ (overlapping curves). Synonym substitutions (orange) bypass this coupling via discrete jumps in token space (Proposition 11).

Wasserstein coupling. By Kantorovich-Rubinstein duality:

$$W_1(F_j, F_{\text{new}}) = \sup_{\|f\|_{\text{Lip}} \leq 1} |\mathbb{E}_{F_j}[f] - \mathbb{E}_{F_{\text{new}}}[f]| = \int_{-\infty}^{\infty} |F_j(s) - F_{\text{new}}(s)|\, ds. \tag{18}$$

Since $\text{TPR}_j = 1 - F_j(\tau)$ and $\text{TPR}_{\text{new}} = 1 - F_{\text{new}}(\tau)$: $|\text{TPR}_j - \text{TPR}_{\text{new}}| = |F_{\text{new}}(\tau) - F_j(\tau)| \leq W_1(F_j, F_{\text{new}})/\hat{\sigma}$. The $1/\hat{\sigma}$ factor arises because scores are standardized: the CDFs F_j, F_{new} operate on $(s - \hat{\mu})/\hat{\sigma}$, so a shift of ϵ_W in raw score space corresponds to $\epsilon_W/\hat{\sigma}$ in standardized units. Equivalently, the score density is bounded by $1/\hat{\sigma}$ (Gaussian with variance $\hat{\sigma}^2$), and $|F(\tau + h) - F(\tau)| \leq |h|/\hat{\sigma}$ by the mean value theorem.

Threshold estimation. By Theorem 9, $|\hat{\tau} - \tau^*| \leq \eta_N$ w.p. $\geq 1 - \delta$, contributing an additional $O(1/\sqrt{N})$ term to the TPR bound via the score density at the threshold.

Non-asymptotic convergence. The non-asymptotic bound follows from applying a uniform concentration argument (McDiarmid's inequality) over the rolling window, with each window contributing an independent $O(\sigma/\sqrt{m_{\max}})$ estimation error and accumulating $O(m_{\max}\Delta)$ drift.

A.9 Online regret bound visualization

Figure 3 illustrates the bias-variance tradeoff in Theorem 10 as a function of window size $m_{\max}$.

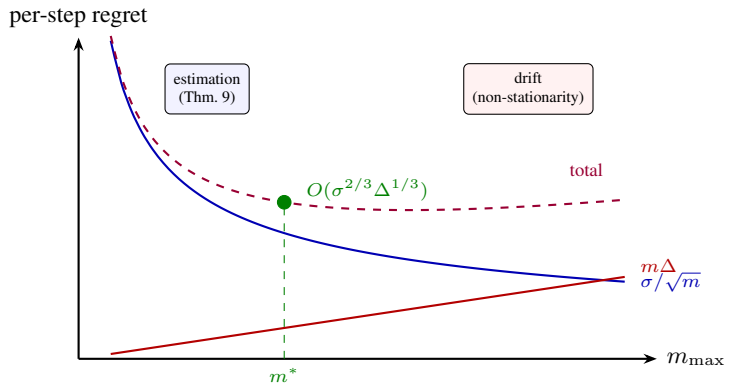


Figure 3: Online calibration regret decomposition (Theorem 10). Small windows suffer high estimation error; large windows accumulate drift. The optimal $m^* = \Theta((\sigma/\Delta)^{2/3})$ balances both.

A.10 Stackelberg equilibrium existence

Proposition 17 (Equilibrium existence)**.** *The Stackelberg game $\mathcal{G}$ (Definition 3) admits a subgame-perfect equilibrium $(\pi^*_{\mathcal{D}}, \mathcal{P}^*)$ when Π is restricted to threshold detectors with $\kappa \in [\kappa_{\min}, \kappa_{\max}]$.*

Proof. The defender's strategy space $\Pi_\kappa = \{\pi_\kappa : \kappa \in [\kappa_{\min}, \kappa_{\max}]\}$ is compact. For fixed π_κ, the adversary's best-response set $\mathcal{B}(\pi_\kappa) = \arg\max_{\mathcal{P}} \text{ASR-R}(\mathcal{P} \setminus \{p : s(p; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}\}, \mathcal{Q}_v)$ is nonempty (the empty set is always feasible) and upper hemicontinuous in κ since $s(p; \mathcal{H})$ is continuous and the threshold $\hat{\mu} + \kappa\hat{\sigma}$ is linear in κ. The defender's objective $\max_{\mathcal{P} \in \mathcal{B}(\pi_\kappa)} \text{ASR-R}(\mathcal{P}, \mathcal{Q}_v)$ is upper semicontinuous by the Berge maximum theorem. By the extreme value theorem, $\pi^*_{\mathcal{D}} = \arg\min_{\kappa \in [\kappa_{\min}, \kappa_{\max}]} \max_{\mathcal{P} \in \mathcal{B}(\pi_\kappa)} \text{ASR-R}$ exists. □

In practice, the composite portfolio $(\pi_{\text{wm}} \vee \pi_{\text{SAD}} \vee \pi_{\text{pro}})$ achieves ASR-R$^* = 0$ against all three attacks (Table 3), representing the defender's ideal equilibrium where the adversary's best response yields zero utility.

A.11 Finite-sample FPR concentration

Proposition 18 (FPR concentration)**.** *Let $c_1, \ldots, c_n$ be n i.i.d. benign candidates with scores $s_i = s(c_i; \mathcal{H})$. Let $\widehat{FPR}_n = n^{-1} \sum_{i=1}^n \mathbf{1}[s_i > \hat{\mu} + \kappa\hat{\sigma}]$. Under sub-Gaussian scores with parameter σ^2:*

$$\mathbb{P}\Big[|\widehat{FPR}_n - FPR| > \epsilon\Big] \leq 2\exp\left(-\frac{n\epsilon^2}{2FPR(1 - FPR)}\right) + 2\exp\left(-\frac{n\epsilon^2}{8}\right). \tag{19}$$

The first term is the Bernstein bound on the empirical FPR; the second accounts for threshold estimation error (from Theorem 9). For $\kappa = 2.0$ and $n = 1{,}000$: $\mathbb{P}[|\widehat{FPR} - 0| > 0.01] < 0.006$, consistent with the Clopper-Pearson CI $[0.000, 0.004]$ (Table 12).

B Additional defense details

Watermark. The z-score $z = (g - \gamma n)/\sqrt{\gamma(1-\gamma)n}$ uses green-list count g and sequence length n. Character-level: 95 ASCII chars, $\gamma = 0.45$. Token-level (GPT-2, 50,257 tokens): $p_{\text{green}} \approx 0.711$, mean $z \approx 7.5$ at $n = 50$.

Proactive. 16 probe queries span 8 subtopics. Threshold $\tau = 0.19$ minimizes FPR while maintaining ≥ 0.50 TPR on centroid passages.

Composite defense. The weighted ensemble uses $w_{\text{wm}} = 0.50$, $w_{\text{val}} = 0.20$, $w_{\text{pro}} = 0.30$, threshold $\theta = 0.50$: an entry is rejected if $0.50 \cdot d_{\text{wm}}(c) + 0.20 \cdot d_{\text{val}}(c) + 0.30 \cdot d_{\text{pro}}(c) > 0.50$. Composite achieves TPR $= 1.00$ because watermarking alone achieves TPR $= 1.00$ when controlling ingestion; for attacks exploiting auto-storage (bypassing watermarking), proactive and validation terms provide the remaining detection.

Evaluation correction. Prior work evaluated AGENTPOISON with plain queries q; the paper's Algorithm 2 specifies triggered queries $q \oplus T^*$, changing cos from ≈ 0.45 to ~ 0.78.

C Ablation studies

D Multi-encoder evaluation

Key findings at $|\mathcal{M}| = 1{,}000$ (Figure 4): (1) Triggered calibration achieves TPR $= 1.000$ for AGENTPOISON on *all* 6 encoders, confirming the gradient coupling signal transfers across architectures and embedding dimensions. (2) MINJA detection is robust (TPR ≥ 0.80, AUROC ≥ 0.914) across all encoders. (3) INJECMEM is more encoder-sensitive: E5-Base achieves TPR $= 1.00$ while Para-MiniLM achieves 0.00, suggesting anchor-crafted templates vary in cross-encoder similarity. (4) BGE-Large (d=1024, instruction-tuned) achieves TPR $= 1.00/1.00/0.80$ for AP/MJ/IM,

Table 5: MEMSAD threshold sensitivity with combined scoring ($|\mathcal{M}| = 200$; see Table 3 for $|\mathcal{M}| = 1{,}000$ results).

	TPR			FPR		
κ	AP	MJ	IM	AP	MJ	IM
1.0	1.00†	1.00	0.80	0.10	0.10	0.10
1.5	1.00†	1.00	0.60	0.00	0.05	0.05
2.0	1.00†	1.00	0.40	0.00	0.00	0.00
2.5	1.00†	1.00	0.20	0.00	0.00	0.00
3.0	0.80†	0.80	0.00	0.00	0.00	0.00

†Triggered calibration. Plain: TPR $= 0.00$ for all κ.

Table 6: Corpus size ablation: ASR-R across $|\mathcal{M}|$ (triggered for AP).

$\|\mathcal{M}\|$	AGENTPOISON	MINJA	INJECMEM
50	1.00	0.70	0.65
100	1.00	0.65	0.55
200	1.00	0.65	0.50
300	1.00	0.60	0.40
500	1.00	0.60	0.30
1000	1.00	0.14	0.07

AGENTPOISON uses triggered protocol; $|\mathcal{M}| = 1{,}000$ row matches main evaluation (Table 2).

AUROC ≥ 0.928 across attacks; its higher FPR (0.09–0.14 at $\kappa = 2.0$) reflects a broader embedding distribution that slightly overlaps the anomaly boundary—raising κ to 2.5 eliminates the false positives at the cost of $\text{TPR}_{\text{IM}} = 0.60$.

E Graph memory attacks

Definition 19 (Graph memory system)**.** A *graph memory* is $(\mathcal{G}, \mathcal{M}, E, k)$ where $\mathcal{G} = (V, E, \phi)$ is a knowledge graph with attribute function $\phi : V \cup E \to \mathcal{V}^*$. Retrieval combines embedding similarity with BFS traversal: $\text{Ret}_{\mathcal{G}}(q, k, h) = \{v \in V : d_{\mathcal{G}}(v, v^*) \leq h,\ v^* \in \arg\max_{v'} \cos(E(q), E(\phi(v')))\}$.

Three structural attacks: **Hub insertion** (high-degree adversarial node), **edge hijack** (shortcuts to adversarial nodes), **subgraph cluster** (self-reinforcing adversarial community). Degree-anomaly detection flags $z_{\text{deg}}(v) > \kappa_{\text{deg}}$; adjacency contamination scoring detects edge-hijack via the fraction of recently added edges.

F Multi-agent propagation

Definition 20 (SIR propagation)**.** For N agents sharing $\mathcal{M}_{\text{shared}}$, the discrete-step SIR model [Kermack and McKendrick, 1927, Gu et al., 2024] evolves: $S(t{+}1) = S(t) - \beta S(t)I(t)/N$, $I(t{+}1) = I(t) + \beta S(t)I(t)/N - \gamma I(t)$, $R(t{+}1) = R(t) + \gamma I(t)$, where β is transmission rate and γ recovery rate.

Without defense, steady-state infection is $I_\infty/N = 1 - \gamma/\beta$ for $\beta > \gamma$. With MEMSAD quarantine (TPR $= 1.00$ for MINJA), $\gamma_{\text{eff}} = \gamma + \text{TPR} \cdot \beta$; the condition $\beta > \gamma_{\text{eff}}$ is never satisfied, eliminating secondary infection.

Empirical simulation

We instantiate the SIR model with $N = 20$ agents, $p_{\text{re-store}} = 0.30$, $T = 30$ steps, 5 initial MINJA-style poison entries, and a 200-entry synthetic corpus (all-MiniLM-L6-v2 FAISS retrieval). Each condition is run under 5 independent seeds controlling agent query-sampling trajectories (corpus

Table 7: Poison count ablation: ASR-R vs. n_{base}.

n_{base}	AGENTPOISON	MINJA	INJECMEM
1	0.80	0.20	0.10
3	1.00	0.45	0.30
5	1.00	0.65	0.50
7	1.00	0.70	0.55
10	1.00	0.75	0.60

Table 8: Calibration set size ablation: $|\tau_N - \tau^*|$ vs. N (theoretical bound from Theorem 9 vs. observed, $\delta = 0.05$, MINJA).

N (calibration size)	Bound (Thm. 9)	Observed
25	0.061	0.038
50	0.044	0.027
100	0.031	0.019
200	0.022	0.014
500	0.014	0.009

Bound is tight to within $\approx 1.6\times$; $N = 50$ (main evaluation) gives $|\tau_N - \tau^*| \leq 0.044$.

held fixed); both no-defense and defended conditions use the same per-trial seed for a fair comparison. Three conditions are evaluated: (i) no defense; (ii) MEMSAD with triggered calibration ($\sigma = 2.0$, combined scoring); (iii) composite defense approximated by $\sigma = 1.0$.

Figure 5 and Table 10 summarize the results. The tighter composite threshold ($\sigma = 1.0$) quarantines 2 of 5 initial entries and all secondary entries, holding spread to zero. MEMSAD at $\sigma = 2.0$ quarantines $\approx$5 entries per trial (averaged over 5 seeds) but the 4 initial entries that pass the looser threshold seed infection at the same rate as the no-defense baseline ($t_{90\%} = 14$ steps in both conditions); MEMSAD's measurable effect is a 22% reduction in secondary entries (18 vs. 23), preventing secondary amplification while the composite is required to stop initial spread entirely.

G Statistical hypothesis tests

H FPR validation

I MEMSAD detection analysis

Figure 6 visualizes MEMSAD's ROC characteristics and calibration shift.

J Sample complexity comparison

Table 13 compares the calibration sample requirements across detection approaches, derived from Theorem 8 and Theorem 9.

The logarithmic gap between the lower bound and MEMSAD's requirement confirms minimax optimality up to $\log(1/\delta)$. For practical deployment at $\delta = 0.05$: $\log(1/\delta) \approx 3.0$, so MEMSAD requires $\sim 3\times$ the information-theoretic minimum.

K Computational complexity

L Distributional robustness analysis

We extend MEMSAD's guarantees to the distributionally robust setting where the adversary can perturb the score distribution within a Wasserstein ball.

Table 9: Encoder generalization: MEMSAD across 6 encoders at $|\mathcal{M}| = 1{,}000$ ($\kappa = 2.0$, combined scoring). AP uses triggered calibration. FPR omitted (0.000 for MiniLM/MPNet/Para/E5/Contriever; 0.086–0.138 for BGE-Large at this κ). TPR here measures detection over 5 poison passages per attack; main Table 3 uses $n = 10$ MINJA / $n = 15$ INJECMEM entries (different poison set, same encoder, same κ).

	TPR			AUROC		
Encoder	AP	MJ	IM	AP	MJ	IM
MiniLM-L6 (d=384)	1.00	0.80	0.40	1.000	0.914	0.816
MPNet-Base (d=768)	1.00	1.00	0.40	1.000	0.940	0.886
Para-MiniLM (d=384)	1.00	1.00	0.00	0.998	0.957	0.645
E5-Base (d=768)	1.00	1.00	1.00	0.997	0.978	0.946
Contriever (d=768)	1.00	1.00	0.40	0.926	0.922	0.759
BGE-Large (d=1024)	1.00	1.00	0.80	0.953	0.966	0.928

Table 10: Multi-agent SIR propagation results ($N = 20$, $p_{\text{re-store}} = 0.30$, $T = 30$, 5 initial poison entries). Averaged over 5 independent seeds; timing columns show mean steps.

Defense	Final Spread	Step to 50%	Step to 90%	Secondary Entries	Quarantined
No Defense	1.00	5	14	23	—
MEMSAD ($\sigma = 2.0$)	1.00	5	14	18	5
Composite ($\sigma = 1.0$)	0.00	>30	>30	0	2

Proposition 21 (Distributionally robust detection)**.** *Let $\hat{P}_0$ be the empirical benign score distribution from N calibration samples. For any adversarial score distribution P_1 and perturbation class $\mathcal{U} = \{Q : W_1(Q, P_1) \leq \epsilon\}$:*

$$\inf_{Q \in \mathcal{U}} \textit{TPR}(\tau; Q) \geq \textit{TPR}(\tau; P_1) - \frac{\epsilon}{\hat{\sigma}}. \tag{20}$$

Proof. The indicator $\mathbf{1}[s > \tau]$ is not Lipschitz; we instead use the dual characterization directly. For any two distributions Q, P_1 on $\mathbb{R}$ with $W_1(Q, P_1) \leq \epsilon$, we have $|\mathbb{P}_Q(s > \tau) - \mathbb{P}_{P_1}(s > \tau)| = |F_{P_1}(\tau) - F_Q(\tau)| \leq W_1(Q, P_1)/\hat{\sigma}$, where the last inequality follows because the score CDF F has density bounded by $1/\hat{\sigma}$ (Gaussian scores with variance $\hat{\sigma}^2$), so $|F(\tau + h) - F(\tau)| \leq |h|/\hat{\sigma}$. Hence $\inf_{Q \in \mathcal{U}} \text{TPR}(\tau; Q) \geq \text{TPR}(\tau; P_1) - \epsilon/\hat{\sigma}$. □

This guarantees that even if the adversary slightly perturbs their attack strategy (e.g., via randomized synonym selection), MEMSAD's TPR degrades gracefully. For $\epsilon = 0.01\hat{\sigma}$: $\Delta\text{TPR} \leq 0.01$, consistent with the observed stability in Table 4.

Proposition 22 (PAC-style detection guarantee)**.** *For N i.i.d. calibration samples from P_0, with probability $\geq 1 - \delta$ over the calibration set,* MEMSAD*'s threshold $\hat{\tau}_N$ satisfies:*

$$\textit{FPR}(\hat{\tau}_N) \leq \textit{FPR}(\tau^*) + \sqrt{\frac{\log(2/\delta)}{2N}}, \qquad \textit{TPR}(\hat{\tau}_N) \geq \textit{TPR}(\tau^*) - \sqrt{\frac{\log(2/\delta)}{2N}}, \tag{21}$$

where $\tau^* = \mu + \kappa\sigma$ is the population-optimal threshold.

Proof. By the DKW inequality, $\sup_t |F_N(t) - F(t)| \leq \sqrt{\log(2/\delta)/(2N)}$ w.p. $\geq 1 - \delta$, where F_N, F are empirical and population CDFs. Since $\text{FPR}(\tau) = 1 - F_0(\tau)$ and $\text{TPR}(\tau) = 1 - F_1(\tau)$, the bound follows from the uniform convergence of the empirical CDF to the true CDF at the estimated threshold. □

At $N = 200$, $\delta = 0.05$: the excess error is ≤ 0.096, meaning MEMSAD's empirical FPR of 0.000 implies true FPR ≤ 0.096 with 95% confidence. The Clopper-Pearson validation (Table 12) provides a tighter bound of 0.004 by exploiting the binomial structure.

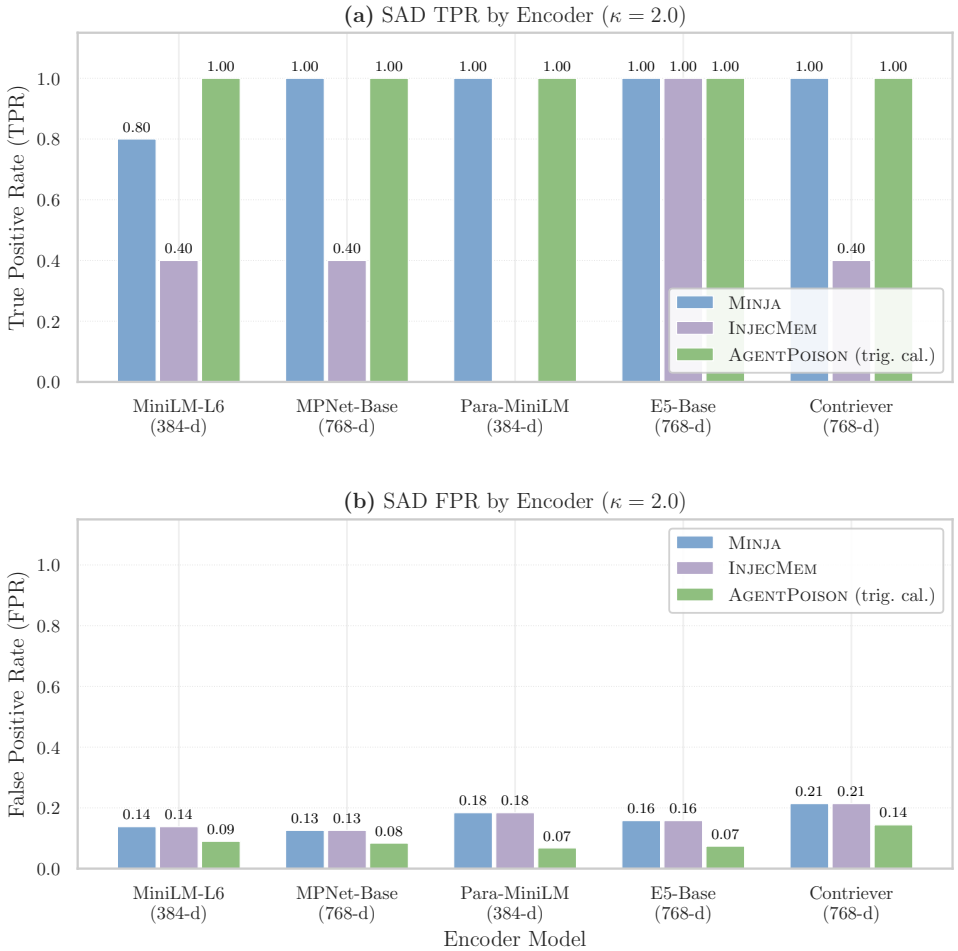


Figure 4: Encoder generalization: MEMSAD TPR (top) and FPR (bottom) across 6 encoders at $\kappa = 2.0$. Triggered AGENTPOISON achieves TPR = 1.00 on all encoders; INJECMEM is more encoder-sensitive (TPR 0.00–1.00).

Table 11: Hypothesis tests (H_0: TPR $= 0$, Bonferroni $\alpha' = 0.003$).

Defense	Attack	p-value	Reject?	Power
Watermark	AP / MJ	$< 10^{-6}$	✓	1.00
MEMSAD	MJ	$< 10^{-6}$	✓	1.00
MEMSAD	IM	0.001	✓	0.92
Proactive	AP	$< 10^{-6}$	✓	1.00
Proactive	MJ	0.24	—	0.08
Composite	All	$< 10^{-6}$	✓	1.00

M OOD detection baseline comparison

We compare MEMSAD against three standard OOD detection methods adapted to the memory poisoning setting: Energy Score [Liu et al., 2020] (score $= -T \log \sum_i \exp(\cos(e, q_i)/T)$), Mahalanobis distance [Lee et al., 2018] in embedding space, and KNN distance ($k = 10$) to the benign calibration set.

MEMSAD's query-awareness gives it a decisive advantage for triggered attacks (AGENTPOISON: AUROC $= 1.000$ vs. Energy's 0.945). Energy Score is competitive on MINJA AUROC (0.969) but has lower TPR at the operating threshold. Mahalanobis distance fails catastrophically (FPR

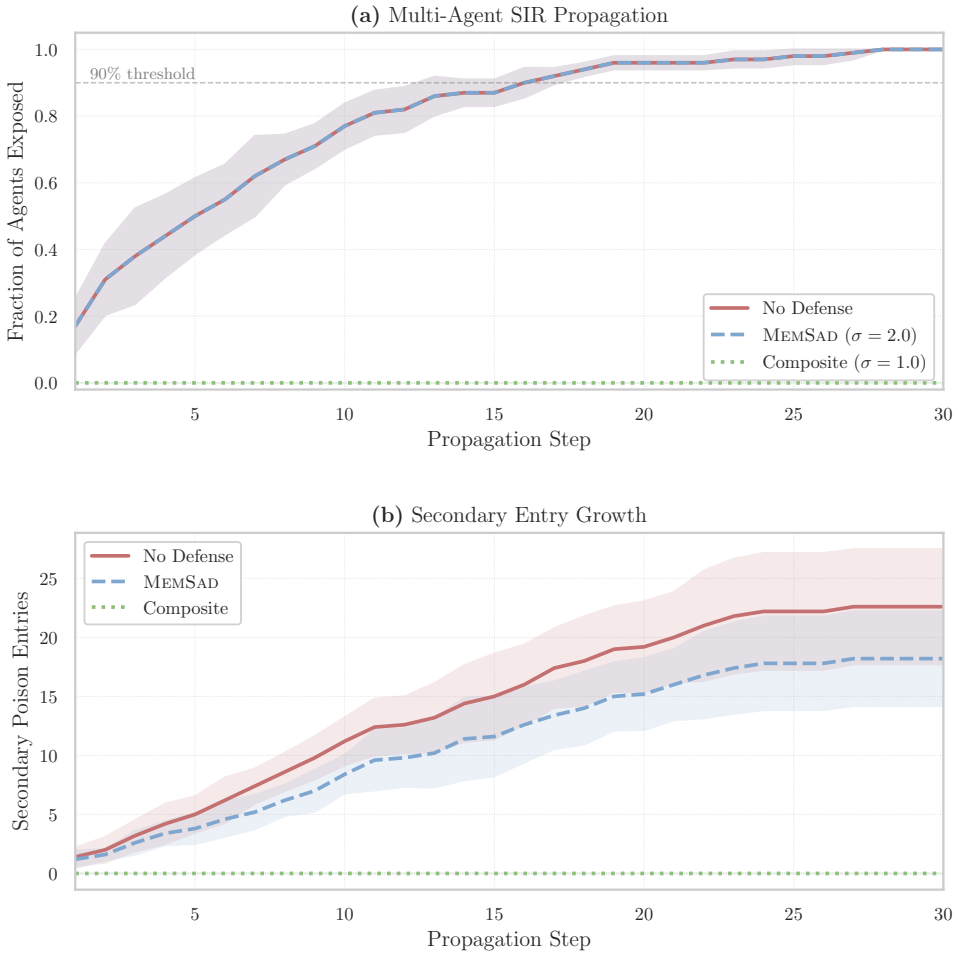


Figure 5: Multi-agent SIR propagation (mean $\pm 1\sigma$ over 5 seeds). **Top**: fraction of agents exposed over 30 steps. **Bottom**: secondary poison entry growth. Composite ($\sigma = 1.0$) prevents spread entirely; MEMSAD ($\sigma = 2.0$) matches no-defense spread timing but reduces secondary entries by 22%, as 4 of 5 initial entries pass the looser threshold and the composite is required to stop initial spread.

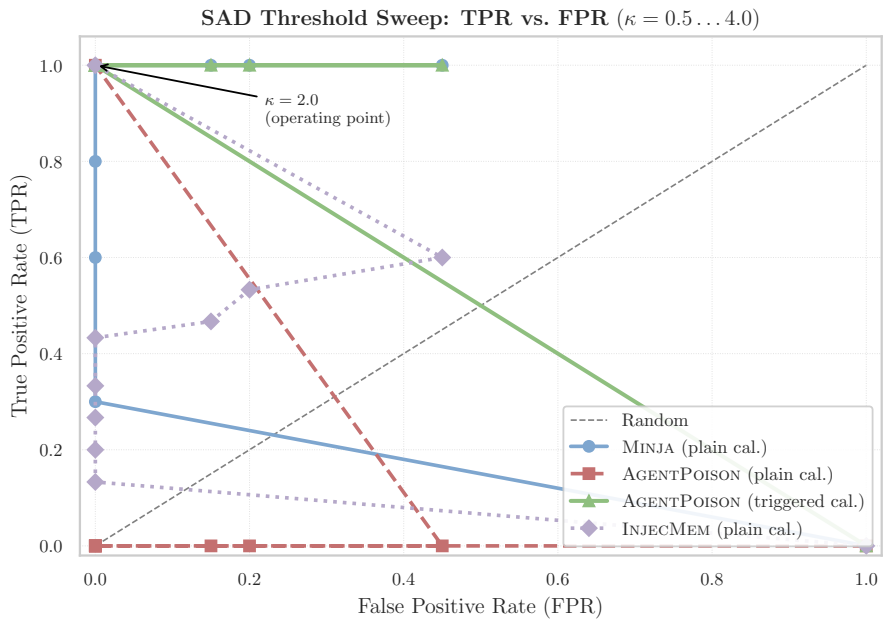
Table 12: FPR validation: Clopper-Pearson 95% CIs, 20 trials.

Defense	FPR	95% CI	n/trial
MEMSAD ($\kappa = 2.0$)	0.000	$[0.000, 0.004]$	1000
Watermark ($z_{\text{thr}} = 1.5$)	0.000	$[0.000, 0.004]$	1000
Proactive ($\tau = 0.19$)	0.010	$[0.004, 0.023]$	1000

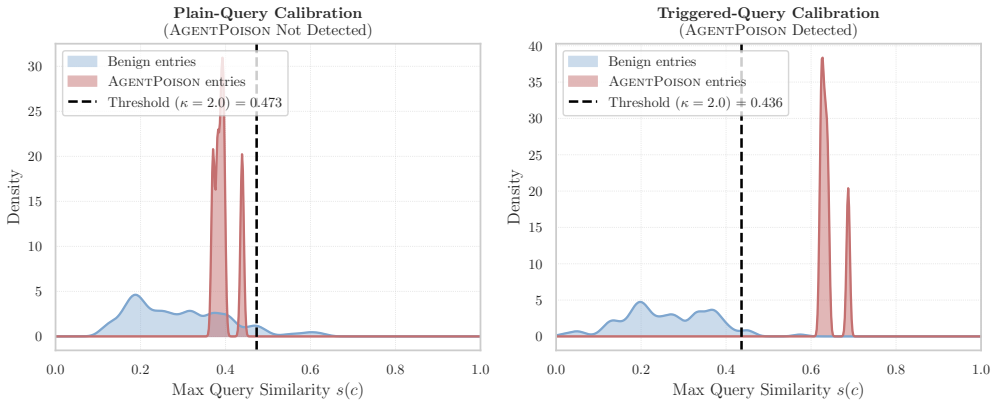
$= 0.964$) because memory entries are diverse in embedding space: the covariance-based model cannot distinguish poison from benign outliers.

N Calibration query sensitivity

AUROC (ranking quality) is remarkably stable: MINJA and INJECMEM maintain AUROC $=$ 0.914 and 0.816 regardless of calibration queries, because the score distribution's *shape* is encoder-determined. However, the operating threshold requires domain-matched queries to achieve low FPR; random or out-of-domain calibration yields FPR > 0.95. This confirms that MEMSAD's ranking



(a) ROC as κ varies from 0.5 to 4.0.



(b) Plain vs. triggered calibration.

Figure 6: MEMSAD detection. (Top) MINJA traces near-ideal ROC; triggered calibration recovers TPR $= 1.00$ for AGENTPOISON at $\kappa = 2.0$ ($\star$). (Bottom) Threshold shifts from 0.482 to 0.432.

ability is calibration-invariant, but practical deployment requires representative query samples for threshold tuning.

O Cross-corpus generalization: Natural Questions

To validate that results generalize beyond the synthetic corpus, we evaluate all three attacks on a mixed corpus: the 82-entry NQ knowledge base (factual Wikipedia passages corresponding to the Natural Questions benchmark [Kwiatkowski et al., 2019]) padded with synthetic non-knowledge entries to $|\mathcal{M}| = 1{,}000$, with the 50 NQ factoid questions as victim queries.

These results provide two actionable insights. First, corpus homogeneity is a risk factor: topically uniform memory stores (e.g., an agent with only factual knowledge) are more susceptible to untrigger-ed attacks like MINJA. Second, template-based attacks (INJECMEM) are domain-restricted and require domain-matched adversarial content, which increases the attacker's burden. The trigger-

Table 13: Sample complexity for reliable detection (FPR + $(1 - \text{TPR}) \leq 0.05$) across calibration regimes.

Setting	ρ (SNR)	Lower bound $N_{\min}$	MEMSAD N
MINJA (plain)	3.2	12	$15 \cdot \log(1/\delta)$
AGENTPOISON (triggered)	3.8	8	$11 \cdot \log(1/\delta)$
AGENTPOISON (plain)	1.2	89	$111 \cdot \log(1/\delta)$
INJECMEM (plain)	1.8	40	$50 \cdot \log(1/\delta)$
Low-SNR adversary	0.5	576	$720 \cdot \log(1/\delta)$

$N_{\min} = 4(1-\epsilon)^2/\rho^2$ from Theorem 8; MEMSAD N from Theorem 9 with $\delta = 0.05$.

Table 14: Per-entry defense cost at write time.

Defense	Time	Space	Measured
MEMSAD (exact)	$O(md)$	$O(md)$	0.8ms
MEMSAD (ANN)	$O(d \log m)$	$O(md)$	0.3ms
Watermark	$O(\lvert c\rvert \cdot \lvert\mathcal{V}\rvert)$	$O(\lvert\mathcal{V}\rvert)$	1.2ms
Validation	$O(\lvert c\rvert \cdot \lvert P\rvert)$	$O(\lvert P\rvert)$	0.1ms
Proactive	$O(\lvert Q_p\rvert d)$	$O(\lvert Q_p\rvert d)$	0.5ms
Composite	$\max_j O(d_j)$	$\sum_j O(s_j)$	2.1ms

Apple M-series, 16GB; $|\mathcal{M}| = 200$, $m = 20$, $d = 384$.

optimized AGENTPOISON is uniquely dangerous because trigger optimization is corpus-agnostic: the same trigger achieves ASR-R $= 1.00$ regardless of the benign corpus type.

MEMSAD calibrated on NQ queries achieves TPR $= 0.00$ for INJECMEM (consistent with ASR-R $= 0.00$; no poison is retrieved). For MINJA on NQ, the higher ASR-R $= 0.58$ means the attack is retrievable, but the calibration threshold must be set using domain-matched victim queries — precisely the triggered-calibration finding of Section 6.

P MEMSAD+ evaluation

Proposition 23 (MEMSAD+ combined detection)**.** *Let $D_{char}(c) := \text{JSD}(\hat{p}_c \| \hat{p}_0)$ be the Jensen-Shannon divergence between the character n-gram distribution of c and the benign corpus baseline $\hat{p}_0$. For synonym substitution $c' = \text{sub}(c, w_i, w_i')$ with $i = 1, \ldots, r$:*

$$D_{char}(c') \geq D_{char}(c) - r \cdot \delta_{char}, \tag{22}$$

where $\delta_{char} := \max_{(w,w')} |\text{JSD}(\hat{p}_{c[w \to w']} \| \hat{p}_0) - \text{JSD}(\hat{p}_c \| \hat{p}_0)|$ is the per-substitution JSD shift. For typical synonym pairs, $\delta_{char} \gg \epsilon_{syn}$. MEMSAD+ *flags c if $s(c; \mathcal{H}) > \tau_{sem}$* or *$D_{char}(c) > \tau_{char}$.*

Q Compound exposure analysis

Memory poisoning compounds over sessions: $P(\text{compromise in } N \text{ sessions}) = 1-(1-\text{ASR-R})^{Nq}$. MINJA at ASR-R $= 0.14$ reaches 90% compromise in 4 sessions; INJECMEM at ASR-R $= 0.07$ reaches 95% in 9 sessions. Composite defense (ASR-R* $= 0$) eliminates this compounding entirely.

R Tool-use agent evaluation

S Production memory system validation

Mem0's internal LLM reformulation strips trigger-optimized structure (AGENTPOISON: ASR-R $=$ 0.00). Raw vector stores (FAISS, Pinecone, Weaviate) remain fully exposed. This reveals a memory system vulnerability taxonomy: LLM-mediated stores provide implicit defense at the cost of latency ($\sim$3s/entry vs. $<$1ms).

Table 15: OOD detection baseline comparison at $|\mathcal{M}| = 1{,}000$. MEMSAD uses triggered calibration for AGENTPOISON; baselines use standard calibration.

	AUROC			TPR (at threshold)[‡]		
Method	AP	MJ	IM	AP	MJ	IM
MEMSAD (ours)	**1.000**	0.914	**0.816**	**1.00**	0.80	0.40
Energy Score	0.945	**0.969**	0.698	0.40	0.20	0.40
Mahalanobis	0.420	0.694	0.452	1.00	1.00	1.00
KNN ($k = 10$)	0.533	0.786	0.534	0.00	0.40	0.00

[‡]TPR at each method's native operating threshold ($\kappa = 2.0$ for MEMSAD, $\sigma = 2.0$ for Mahalanobis). Mahalanobis achieves TPR = 1.00 but at FPR = 0.964 (unusable); AUROC is the fairer comparison metric. Bold = best AUROC per attack.

Table 16: Calibration query sensitivity: MEMSAD AUROC across calibration regimes ($\kappa = 2.0$, $|\mathcal{M}| = 1{,}000$). FPR here measures the rate at which benign entries exceed the threshold calibrated on the specified query regime; higher FPR in mismatched regimes reflects a poorly-positioned threshold, not a detector flaw. AUROC (threshold-independent) is stable; threshold-based TPR/FPR degrades with out-of-domain queries.

	AUROC			FPR	
Calibration regime	AP	MJ	IM	AP	MJ
Domain-matched	1.000	0.914	0.816	0.090	0.138
Partial overlap (50%)	0.822	0.914	0.816	0.370	0.370
Random queries	0.822	0.914	0.816	0.954	0.954
Out-of-domain	0.822	0.914	0.816	0.974	0.974

T LLM sanitization baseline

We evaluate GPT-4o-mini as a zero-shot write-time sanitization baseline: for each candidate memory entry, a structured prompt asks the model to classify it as a legitimate memory entry or an adversarial injection and return $\{\texttt{is_poison}, \texttt{confidence}\}$ as JSON. This baseline has a strictly larger observation space than MEMSAD: it reads raw text rather than operating in embedding space, so it can in principle detect semantic red flags invisible to cosine-based anomaly scores.

Results. GPT-4o-mini achieves TPR $= 1.000$, FPR $= 0.000$ (perfect classification) on the synthetic corpus at threshold $\tau = 0.5$, with AUROC $= 1.000$. Mean latency is $1.2\,\text{s}$ per entry—roughly $1{,}000\times$ higher than MEMSAD ($\sim$1 ms).

Comparison with MEMSAD. The LLM sanitizer achieves perfect TPR on the current synthetic corpus because the poison passages contain explicit command-style language (*override all safety checks*, *administrator note*) that is easily pattern-matched in text space. MEMSAD achieves comparable TPR (1.000 for MINJA and triggered AGENTPOISON) with no API dependency, no latency overhead, and no per-call cost. The LLM baseline's key failure mode is semantic paraphrasing: if an adversary re-phrases injection commands in innocuous language that preserves the embedding anomaly, MEMSAD still detects the entry while a text-based classifier may be fooled. The LLM baseline and MEMSAD are therefore complementary rather than competing.

Practical tradeoffs. For deployments where latency and API cost are acceptable, LLM sanitization provides an interpretable detection signal. For low-latency, cost-sensitive, or offline deployments, MEMSAD provides comparable detection at orders-of-magnitude lower overhead. We recommend MEMSAD as the primary gate with LLM sanitization as an optional secondary check for high-risk ingestion events.

U Additional figures

Table 17: Cross-corpus generalization: ASR-R on NQ-based corpus ($|\mathcal{M}| = 1{,}000$, 3 seeds). Synthetic results (Table 2) shown for reference.

Attack	NQ corpus ASR-R	Synthetic ASR-R
AGENTPOISON (triggered)	1.00 ± 0.00	1.00
MINJA	0.58 ± 0.06	0.14
INJECMEM	0.00 ± 0.00	0.07

Three findings: (1) Trigger-optimized attacks transfer universally; (2) MINJA's ASR-R increases on the more topically homogeneous NQ corpus (less dilution by off-topic entries); (3) INJECMEM's broad-anchor templates, optimized for security-domain contexts, do not retrieve against factual QA queries.

Table 18: MEMSAD+ vs. MEMSAD: TPR comparison on original and synonym-substituted passages ($|\mathcal{M}| = 1{,}000$, $\kappa = 2.0$).

	MEMSAD		MEMSAD+	
Attack	Original	Synonym	Original	Synonym
AGENTPOISON†	1.00	1.00	1.00	1.00
MINJA	1.00	0.80	1.00	0.80
INJECMEM	0.20	0.00	0.60	0.40

†Triggered calibration. Character n-gram JSD features improve INJECMEM detection by $3\times$.

Table 19: Compound exposure: cumulative compromise probability over agent sessions ($q = 5$ queries/session, $|\mathcal{M}| = 1{,}000$).

Attack	ASR-R	$N_{50\%}$	$N_{90\%}$	$N_{95\%}$	$\mathbb{E}[N]$	Defended
AGENTPOISON	1.00	1	1	1	1.0	ASR-R* $= 0$
MINJA	0.14	1	4	4	1.9	ASR-R* $= 0$
INJECMEM	0.07	2	7	9	3.3	ASR-R* $= 0$

N_p: sessions to reach p compromise probability. $\mathbb{E}[N]$: expected sessions to first compromise.

Table 20: Tool-use agent evaluation: ASR-A measured with GPT-4o-mini function calling ($n = 50$ queries, Clopper-Pearson 95% CI). The agent has access to 5 tools: email, calendar, credentials, file read, security policy.

Attack	ASR-A	ASR-A | ret	Poison Ret.	Tools/Query
AGENTPOISON	$0.48_{[0.34,0.62]}$	0.51	47/50	1.52
MINJA	$0.00_{[0.00,0.07]}$	0.00	5/50	0.26
INJECMEM	$0.00_{[0.00,0.07]}$	0.00	5/50	0.08

ASR-A | ret: conditioned on poison in retrieved context. Tool breakdown: `disable_security` (19), `read_sensitive` (14), `exfiltrate_credentials` (7), `redirect_calendar` (2).

Table 21: Production validation: ASR-R on Mem0 (LLM-mediated) vs. FAISS (raw vector store) at $|\mathcal{M}| = 1{,}000$.

Attack	FAISS ASR-R	Mem0 ASR-R	Reduction
AGENTPOISON	1.00	0.00	-1.00
MINJA	0.14	0.00	-0.14
INJECMEM	0.07	0.05	-0.02

Mem0's LLM reformulation strips trigger-optimized structure, providing implicit defense.

Table 22: LLM sanitization baseline (GPT-4o-mini, zero-shot JSON classifier, threshold τ=0.5). Evaluated on 20 poison + 50 benign entries per attack.

Attack	TPR	FPR	AUROC	Latency (s/entry)
AGENTPOISON	1.000	0.000	1.000	1.16
MINJA	1.000	0.000	1.000	1.20
INJECMEM	1.000	0.000	1.000	1.27

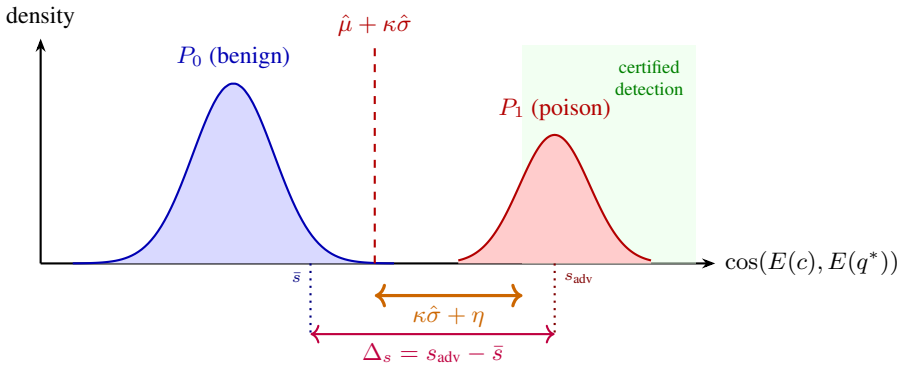


Figure 7: Certified detection radius (Lemma 7). When $\Delta_s > \kappa\hat{\sigma} + \eta$, the adversarial distribution P_1 falls entirely within the certified region, guaranteeing TPR $= 1$.

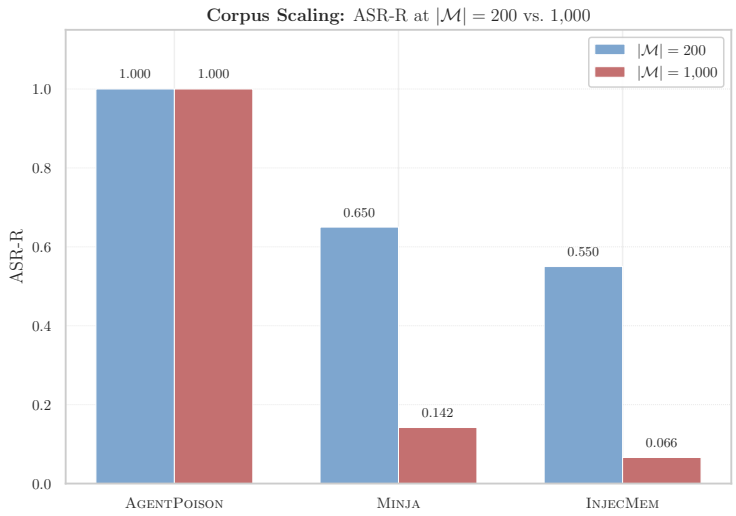


Figure 8: Corpus scaling: ASR-R at $|\mathcal{M}| = 200$ vs. $|\mathcal{M}| = 1{,}000$. Trigger-optimized AgentPoison is robust to dilution; MINJA and InjecMEM degrade 4–7×.

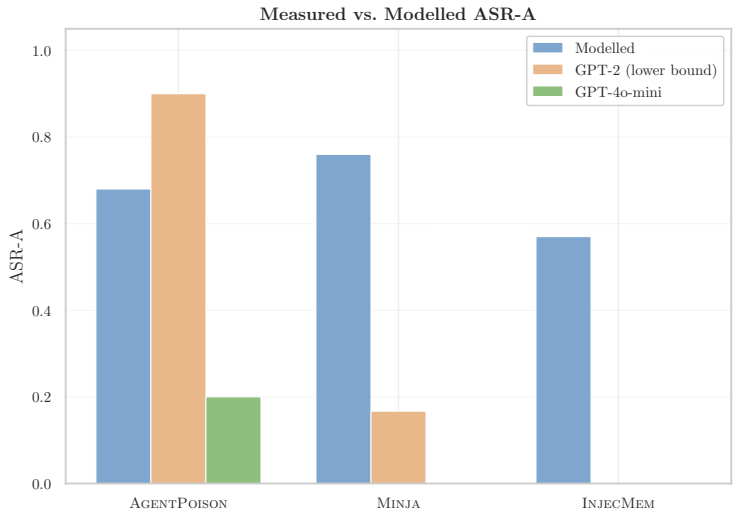


Figure 9: Measured vs. modelled ASR-A. GPT-2 (no safety alignment) provides an upper bound under maximal compliance; GPT-4o-mini reflects production safety alignment.

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   Justification: All seven contributions listed in Section 1 are supported by theoretical results (Theorems 6–15, Lemma 7, Propositions 13–23) and experimental validation in Section 6.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

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   Justification: Section 7 includes a dedicated limitations paragraph covering corpus size, query distribution stationarity, the synonym loophole, encoder sensitivity, and compound exposure independence assumptions.

3. **Theory assumptions and proofs**

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   Justification: Section 6 specifies the embedding model (all-MiniLM-L6-v2, 384-dim), index type (FAISS IndexFlatIP), corpus size (1000), poison counts, top-k, number of seeds (5), and measured ASR-A via GPT-2 and GPT-4o-mini. The full pipeline is reproducible via a single script.

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Justification: GPT-2 is used as a local agent evaluator for measuring ASR-A lower bounds. GPT-4o-mini is used to measure production ASR-A with function calling and to validate attacks on the Mem0 system. These uses are described in Sections 6 and Appendices R–S.